Short brief

Astra Integrasi Digital (AID), part of Astra Financial Group, is a digital intelligence, AI-powered analytics, and connectivity solution provider delivering end-to-end services that accelerate strategic advancement. We help organizations manage, optimize, and leverage their digital intelligence and AI-powered analytics ecosystem to create scalable and reliable solutions. Our capabilities include intelligence management, advanced analytics, AI-powered analytics, connectivity, and customer insight. By combining technology, analytics, artificial intelligence, and industry expertise, AID enables insight-driven decision-making and supports sustainable business growth.

Study Case - Data Analyst

Business Context

You are working at a digital financing company providing lending services.
The business aims to:
- Reduce customer churn (inactive or no repeat borrowing)
- Improve customer retention and repeat usage
- Optimize acquisition channels

You are given datasets containing user profiles, activity, transactions, and churn status.

Dataset Description

1. users: user_id, signup_date, acquisition_channel, age, city
2. activity: user_id, event_date, session_count, feature_used
3. transactions: user_id, transaction_date, amount
4. status: user_id, is_churn (1 = churn, 0 = active), churn_date

Part 1 – SQL (Data Exploration)

1. Calculate total users, active vs churn users, and churn rate.
2. Calculate churn rate by acquisition channel and signup cohort.
3. Identify users who have not logged in for the last 30 days but were previously active.
4. Identify top 3 acquisition channels with best retention and highest churn.

Part 2 – EDA & Insight

1. Compare behavior between churn and non-churn users.
2. Analyze session frequency, transaction amount, and feature usage.
3. Identify characteristics of users likely to churn.
4. Create simple segmentation: high-risk vs low-risk users.

Provide insights and hypotheses, not just charts.

Part 3 – Modeling

1. Build a churn prediction model (Logistic Regression or Tree-based).
2. Explain feature selection.
3. Evaluate model performance (Accuracy, Precision, Recall, AUC).
4. Interpret key drivers of churn.

Part 4 – Business Thinking

1. Suggest 3 strategies to reduce churn.
2. Should the company prioritize retention or acquisition? Explain.
3. With limited budget, which user segment should be targeted?
4. What additional data would improve your analysis?

Part 5 – Deliverables

1. Presentation (max 5–7 slides)
2. Include key findings, recommendations, and business impact
3. Formatting requirements: Calibri font, size 28 for titles, 24 for subtitles, and 12 for body text
4. Kindly submit the presentation materials no later than 3 x 24 hours after this email is sent

In addition, please create a self-introduction video with a maximum duration of 60 seconds, covering your background, personal profile, and experiences that you can contribute to AID. The video can be in Bahasa Indonesia or English and should be uploaded to YouTube (set as Unlisted), Google Drive, or attached directly to this email.

Furthermore, if you pass this stage, we will contact you to proceed to the user interview stage. Thank you.